

MC214: Operating Systems (3-0-2-4) Autumn 2025-26

Instructor: Anish Mathuria

Lectures: TBA

Practicals: TBA

Prerequisites: basic data structures, computer organization

Objectives

This course provides an understanding of the fundamental structures and mechanisms used by an operating system.

Topics to be covered

- Multi-programming: processes and threads, system calls, context switching, types of scheduling, scheduling metrics, scheduling policies
- Concurrency and synchronization: race conditions, critical sections, mutual exclusion, Peterson's algorithm, hardware support for locking, spin lock, ticket lock, blocking lock, semaphores, condition variables, implementation issues, deadlocks, lock-free data structures
- Virtual memory: logical/physical address spaces, paging, protection, TLB, demand paging, copy-on-write, page replacement policies, implementation issues
- File systems: hard-disk drives, device drivers, files and directories, hard links, symbolic links, file system implementation, i-nodes, file layout approaches, crash recovery, journaling, fast file system, solid-state drives, file systems for SSDs

Learning outcomes

At the end of the course, students will have an understanding of

- the main principles and techniques used to implement processes and threads as well as the different algorithms for scheduling;
- the problems related to concurrency, the basic synchronization mechanisms available as well as the techniques used to implement them;
- the data structures and components (both hardware and software) that are used to implement virtual memory;
- how data structures of a file system serve to map file names to blocks of a storage device.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X	X		X				X			

Textbook

Operating Systems: Three Easy Pieces, Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, November 2023 (Version 1.10). The course text covers most of the lecture material. It is freely available at <http://pages.cs.wisc.edu/~remzi/OSTEP/>

References

- Operating System Concepts, A. Silberschatz, P. Galvin and G. Gagne, 8th edition, 2009, Pearson Education.
- Linux Kernel Development, Robert Love, 3rd edition, 2010, Pearson Education.

Evaluation

Two in-semester tests and a final exam will take place at the officially scheduled exam time for the course.

- 30% two mid-term exams
- 40% final exam
- 20% in-lab exercises
- 10% mini-projects

Lab policy

- Lab attendance is mandatory. You will have to demonstrate your code for the in-lab exercises to a TA as specified in the lab handout. If you fail to do so, you will lose points. There will be no makeup labs for any reason.
- It is a course requirement that you do the programming assignments on Linux OS. If the code you submit does not compile on Linux or we are unable to run the executable, you will not receive credit.